



2.6 Algebraic Methods

91261 Greenbook



snedmaths



INTRODUCTION

This workbook contains practice external examinations for Level 2 NCEA:

2.6 91261 Apply algebraic methods in solving problems

Assessment schedules are online at

<https://sites.google.com/view/snedmaths>

©2019 - Jamie Sneddon

Schools and teachers can use this resource however they wish, with the provisos that:

- it is unmodified and retains the link to the site;
- no part of this resource is re-used as part of a site that requires a subscription or fee; and
- no part is used or modified to be a school “derived grade” or “end of topic” examination.

If you find errors, please let me know!

jamie.sneddon@saintkentigern.com

The cover photo is a New Zealand tīeke or South Island saddleback (*Philesturnus carunculatus*). The photo was taken by photographer Bernard Spragg and is in the public domain.

ALGEBRAIC METHODS 91261-EXM-1

QUESTION ONE

- (a) Simplify $(32a^{1.25})^{0.8} \times (a^3)^2$
- (b) Fully simplify

$$\frac{9 - x^2}{x^2 + x - 12}$$

- (c) The volume of a bucket can be calculated from the top radius R , bottom radius r and height H using the equation

$$V = \frac{\pi}{3}h(R^2 + Rr + r^2)$$

- (i) Show that this formula gives the volume of a cylinder when $R = r$.
- (ii) A particular bucket has $r = 0.8R$. If the bucket holds 10 000 cm³ of water, find the height of the bucket h in terms of r .

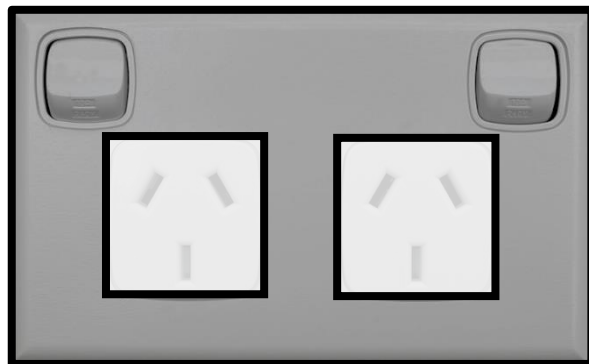
Use this to find the height of the bucket when the upper radius is 11 cm.



- (d) A double socket power point (shown) is $2x + 5$ cm long and $x + 4$ cm wide, where each socket is a square x cm on the side.

The combined area of the two squares is 23% of the total area of the double socket power point.

Calculate the dimensions of the double socket power point.



QUESTION TWO

- (a) Solve the following equations for x :
- (i) $\log_x 7 = 3$
 - (ii) $10^{7x} = 100^{x+1}$
 - (iii) $(\log x)^2 - \log x = 0$
- (b) Determine the values m for which the equation $mx^2 + mx + 1 = 0$ has no solutions.
- (c) Find a simplified expression for x in terms of q :

$$2 \log x + \log q = \frac{1}{3} \log \frac{x}{q}$$

QUESTION THREE

- (a) Simplify

$$\frac{1}{x} - \frac{1}{x+1}$$

- (b) The equation $4x^3 + kx^2 + 35x = 0$ has roots $x = 0$, $x = 2.5$ and $x = 3.5$.
Find the value of k .
- (c) The capital city of the Philippines is Manila. Its population is estimated to grow at a rate of 0.44% each year. The population in 2010 was 10.7 million.
- (i) Write a model for the population of Manila t years after the year 2010 in the form $P(t) = Ab^t$.
 - (ii) The amount of wastewater that Manila produces is proportional to the city population.
Wastewater processing operated at 90% capacity in 2010.
If the amount of wastewater that can be processed each year is increased by only 0.33%, calculate how long it will take before Manila cannot process its wastewater.
- (d) For what value of K does the equation $(Kx)^2 + 11x + K = 0$ have exactly one solution?

ALGEBRAIC METHODS 91261-EXM-2

QUESTION ONE

- (a) Simplify

$$\sqrt[3]{\frac{64a^{12}}{125b^6}}$$

- (b) Simplify $\log(b^2 - b - 6) - \log(b - 3)$, writing as a single logarithm.
- (c) The volume of a truncated pyramid, as shown, is given by the formula

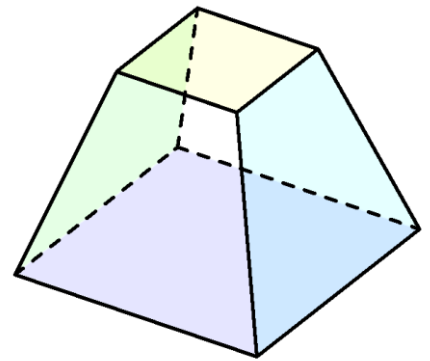
$$V = \frac{h}{3} \left(\frac{b^3 - a^3}{b - a} \right)$$

where

a is the side length of the top square,

b is the side length of the bottom square, and

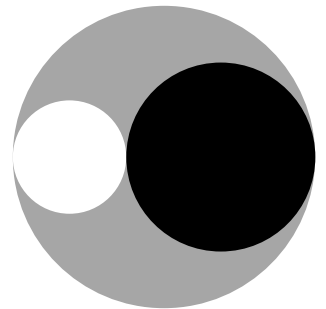
h is the vertical height.



Show that $(b - a)(b^2 + ab + a^2) = b^3 - a^3$ and hence find the volume when $a = b$.

- (d) A logo is made inside a circle of radius 2 cm. It shares a horizontal diameter with two smaller circles. The white circle has radius x cm, where $0 < x < 2$.

Find expressions for the areas of the white, black and grey areas, and hence find when the total grey area and black circle have equal areas.



- (e) Fully simplify and hence solve

$$\frac{x^2 - x - 20}{(x^2 - 16)(x^2 - 25)} = -\frac{4}{81}$$

QUESTION TWO

- (a) Find the value(s) of x .

(i) $\log_{16} x = 2.25$

(ii) $A^{2x} = A^{x^2}$

(iii) $\log x - \log(x - 1) = 2$

- (b) Suppose $x > y > 0$.

Show that $A = 2xy$, $B = x^2 - y^2$ and $C = x^2 + y^2$ are side lengths of a right-angle triangle: that is $A^2 + B^2 = C^2$.

- (c) The equation $Bx^2 - Bx = 126$ has the solution $x = 10$.

Find the value of B and the other solution.

QUESTION THREE

- (a) Expand and simplify $(x + 2)(x - 2)(x^2 + 4)$
- (b) Determine the values of M for which the following quadratic has no solutions.

$$(M + 2)x^2 + (M + 1)x + M = 0$$

- (c) A slow-release block of fish food slowly dissolves in a fish tank. The initial weight of the block is 46 grams.

At the end of each day, 20% of the remaining block has dissolved into the tank.

- (i) Write an equation in the form $f(t) = Kb^t$ for the amount of fish food remaining after t days.

Explain the meaning of the parameters K and b .

- (ii) During one day, the block of fish food reduces in weight by 3 grams. Which day is this?

- (d) Simplify and solve for x :

$$\log(x^2 - (99k)^2) - 2 \log(x - 99k) = 2$$

ALGEBRAIC METHODS 91261-EXM-2

QUESTION ONE

- (a) Simplify $(-4x^3)^{-2}$
- (b) Find the value of $\log_{b^4}(b^7)$.
- (c) Solve to find x in terms of k . $2\sqrt{kx} = \sqrt{x + k}$
- (d) Simplify and hence solve:

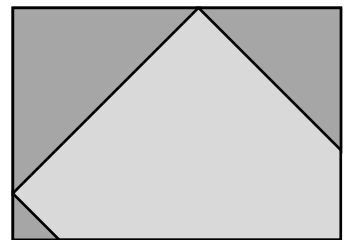
$$\frac{3(x^2 - 3x + 2)}{x^2 - 5x + 6} = x + 1$$

- (e) Three isosceles triangles are removed from the corners of a sheet of paper with sides length L and $1.5L$ centimetres as shown (not to scale).

The triangle at the bottom left has short side lengths x centimetres.

Calculate the area of the removed triangles in terms of x and L .

Simplify your answer.



QUESTION TWO

- (a) Simplify the following, leaving your answer with positive indices:

$$\left(\sqrt{16b^9x^{-2}}\right)^{-1}$$

- (b) Solve the equation $x^2 + kx + k^2 = k^3$ in terms of k , and hence find the value(s) of x when $k = 13$.
- (c) Show that $\log(A^{\log B}) = \log(B^{\log A})$.
- (d) A rectangular prism is made from wooden blocks. Its dimensions are $x + 7$ blocks wide, $x + 5$ blocks long and $x + 1$ blocks tall.

These blocks can be reformed into two cubes: one with side length $x + 4$ blocks and one with side length 3 blocks.

Find the dimensions of the original prism.

- (e) Solve $x^{(x-4)^2} - x^4 = 0$.

QUESTION THREE

- (a) Rearrange to make x the subject of the equation:

(i) $2^{x^2+4k^2} = 4^{2kx}$

(ii) $a \log x^a + b \log x^b = 1$

- (b) The speed of a falling object is modelled by the following equation, where t is the time in seconds after measurements start and v is the speed in metres per second.

$$v_1(t) = 5 \log(t + 3)$$

- (i) Find when the speed is 4 m/s.
- (ii) Another object is falling at a speed of $v_2(t) = 10 \log(t + 1)$.

Find when the two objects have the same speed.

- (c) Show that the following four expressions are equivalent.

$$x^6 = y^6$$

$$(x^3 - y^3)(x^3 + y^3) = 0$$

$$(x^2 - y^2)(x^4 + x^2y^2 + y^4) = 0$$

$$(x - y)(x^5 + x^4y + x^3y^2 + x^2y^3 + xy^4 + y^5) = 0$$

ALGEBRAIC METHODS 91261-EXM-4**QUESTION ONE**

- (a) Find the value of $\log_9 27$.
- (b) Solve the equation $\log_{10}(x^2 - 3x) = 1$.
- (c) The price of a new car stereo was \$379 but needs to be increased by a percentage.

Two staff members have both increased the price of the stereo by this percentage, and the new *incorrect* price is \$466.97 after rounding.

Find the percentage increase both staff members applied, and the correct new price.

- (d) Solve for x .

$$\log_2(x^2 + 2) - \log_2(x - 1) - \log_2(x + 1) = 1$$

- (e) The area of any isosceles triangle with side lengths x , x and y is given by

$$A = \frac{1}{4} \sqrt{4x^2y^2 - y^4}$$

Show that this area is undefined if $y > 2x$ and give a geometric interpretation of this result.

QUESTION TWO

- (a) Write $x^2 - 12x + 2$ in the form $(x - r)^2 + h$.
- (b) Show that if n is a positive integer then the equation $(x - n^2)^2 = n^2$ has two integer solutions.
- (c) Show that the quadratic equation given below has two solutions that depend on the value of A .

$$(A + 1)x^2 + 3Ax + A - 1 = 0$$

- (d) The following curves intersect at the point $(x, y) = (1, 2)$.

$$y = x^2 - 2x + 3$$

$$y = 2x^2 - 6x + 6$$

$$y = 3x^2 - 10x + 9$$

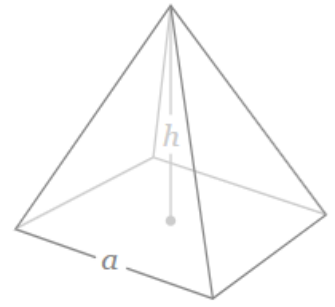
Find the other common point of intersection.

- (e) The sloped surface area of a right square pyramid as shown is given by the formula

$$A = 2a\sqrt{0.25a^2 + h^2}$$

Find an equation for h in terms of A and a .

Hence find the value(s) of a that for a pyramid with sloped surface area 100 cm^2 and height 10 cm .



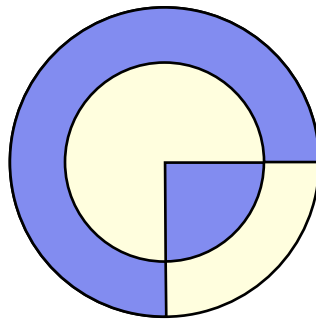
QUESTION THREE

- (a) Simplify each of the following, writing with positive indices.

(i) $(A^2 + 1)^{-1} - A^{-2}$

(ii) $b^{-1}\sqrt{(b + c^{-1})^2}$

- (b) The diagram below shows the icing on a circular cake with radius 14 cm . The outer band of dark blue icing is $x \text{ cm}$ wide, where $0 < x < 7$.



- (i) Show that the area of the blue icing is

$$B = \pi \left(49 + 14x - \frac{1}{2}x^2 \right)$$

- (ii) Find both solutions of $B = 98\pi$, and explain what these solutions mean in the context of the iced cake.

- (c) A maths teacher with a mean streak asks his class to evaluate

$$A = \sqrt{11 + \sqrt{11 + \sqrt{11 + \sqrt{11 + \dots}}}}$$

He gives his class the hint that the number must also satisfy the following equation.

$$A = \sqrt{11 + A}$$

Find the value of A .